Aligning Latent Evolution in Multi-Agent Communication

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Abstract

Multi-Agent Reinforcement Learning (MARL) typically improves coordination through individual weight updates. Taking inspiration from cultural evolution in human populations, we investigate a different axis of adaptation: **communicating multi-agent systems changing through selection over latent communication strategies**, known abstractly as 'memetic evolution'. We formalize communicated latents as transmissible strategies whose frequencies evolve under a learned transmission operator, and we define the central objective of Reward-Aligned Latent Evolution (RALE): a measure of how much the communication patterns that spread are also those that improve joint return. We instantiate this framework in Attention-guided Latent Evolution in Communication (ALEC), which applies an optimizer (ES or MARL) over the attention-based communication graph and introduces decentralized latent memory via GRUs. The proposed method is competitive with MAPPO in MPE2 Predator-Prey, VMAS Discovery/Transport, and StarCraft MultiAgent Challenge v2. We measure alignment between latent replication success and downstream reward in ALEC using RALE and see that, with both ES and RL-based ALEC, higher-reward latents become more probable. Code available at *https://github.com/AlmondGod/latent-evolution*.

1 Introduction and Motivation

In biological evolution, genes are selected over via differences in their host organisms' replication rates. Evolutionary strategies, the analog of biological evolution for neural network optimization, has shown impressive near-linear scaling qualities as parallel workers are added within steps (Salimans et al., 2017).

Evolutionary selection over units of information inhabiting human memory, memes, has been the defining axis of growth (Dawkins, 1976) for the improvement of intelligence over populations of humans. Since ideas can spread in parallel between generations, evolution over ideas is faster than biological evolution, with cycles in days rather then decades (Boyd and Richerson, 1985). This 'memetic' or cultural evolution is akin to Popper's evolutionary epistemology: the frontier of knowledge progresses through variation, error-correction, and selective retention of conjectures (**?**).

Multi-Agent Reinforcement Learning settings, such as populations of agents coordinating to automate knowledge discovery or engineering research, are systems in which communicative objects behave like memes. Such a system may be able to use the learned biases of internet-scale pretraining, RLHF, and communication to narrow the evolutionary search space and "evolve ideas, rather than biologies" (Ouyang et al., 2022; Gan and Isola, 2026).

Critically, once such a system is created (when the dual criterion of memory and communication are satisfied in deployed AI populations), we have little guarantee that the welfare of individual ideas (prevalence in the population) will be aligned to goals of individual hosts (communicating agents). Modern human systems are an example of this: ideas which spread widely are often harmful to their hosts, simply because the ideas that are popular are selected for virality, not fitness improvement for

the host (Boyd and Richerson, 1985; Richerson and Boyd, 1984). Thus, models aligned at deployment time may likely, through interaction with populations of agents, become misaligned. Can we measure and ensure alignment of meme and agent reward, such that as memes emergently spread, agent reward increases?

2 Preliminaries

2.1 Markov Decision Process Background

Our problem setting is the Multi-Agent Partially-Observable Markov Decision Process (Dec-POMDP) (Bernstein et al., 2002). Dec-POMDPs are defined by the tuple

$$\left(S, \{A_i\}_{i=1}^N, P, R, \{O_i\}_{i=1}^N, O, \gamma\right), \tag{1}$$

where each agent i of N recieves observation $o_i \in O_i$ via O, selects action $a_i \in A_i$, and the environment transitions according to combined actions $P : (s, a) \to s'$, yielding reward R for all agents, who aim to maximize the discounted returns $E[\sum_{t=0}^{T} \gamma^t R_t]$ discounted by γ (Littman, 1994). In such partially observable settings, each agent receives only a local observation and may need communication to recover information relevant to coordinated action. Dec-POMDPs are difficult to optimize because of the combinatorial growth of joint action spaces, partial observability, and the non-stationarity induced by simultaneously learning agents (Bernstein et al., 2002).

2.2 Multi-Agent Reinforcement Learning

Reinforcement learning is one method of learning MDP-solving policies by performing policy iteration (evaluating the "quality" of each state via a Value or Q function) and policy improvement (choosing the policy which takes a V/Q-maximizing action). Solutions to Dec-POMDPs began with Nash Q-learning (Hu and Wellman, 2003). Modern deep MARL methods commonly use centralized training with decentralized execution (CTDE), where agents act from local observations but are trained using centralized critics or joint information.

MADDPG (Lowe et al., 2017) adapts DDPG (Lillicrap et al., 2015) to Multi-agent settings using joint-action critics and decentralized actors, founding the Centralized training, decentralized execution (CTDE) tradition of modern multi-agent algorithms. MAPPO (Yu et al., 2021; Schulman et al., 2017), multi-agent learning in continuous action spaces, uses PPO with a centralized value function. MAPPO reinforced the trend that RL in multi-agent settings can be surprisingly effective with few adaptations.

2.3 Multi-Agent Communication

In partially observable settings, each agent receives only a local observation while sharing a collective objective. Coordination therefore often requires information that no single agent can infer alone. Multi-agent communication architectures address this by allowing agents to send messages between policy steps.

RIAL treats communication as additional discrete actions, while DIAL makes the communication channel differentiable during training (Foerster et al., 2016). IC3Net augments each agent with a message and a binary communication gate $g_i^t \in \{0, 1\}$, so that only active broadcasters contribute to the communication pool (Singh et al., 2018).

TarMAC introduces targeted communication via learned soft attention over sender messages. Each agent maintains a recurrent hidden state h_i^t, computes keys, values, and queries, and selectively aggregates messages from other agents (Das et al., 2019). This family of architectures is especially relevant for our setting because it provides a natural interface between communication content and transmission structure.

2.4 Evolutionary Strategies

Evolutionary strategies (ES) are black-box optimization methods that perturb parameters, evaluate the perturbed candidates, aggregate scalar fitness values, and estimate an update direction (Salimans

et al., 2017). One of ES's defining strengths is near-linear gains from scaling parallelized workers since workers only need to communicate scalar fitness values, so it is attractive in distributed settings where synchronization cost is a bottleneck.

Recent work has improved the scalability of ES to much larger models. Evolution Guided GeneRal Optimisation via Low-rank Learning (EGGROLL) structures perturbations as low-rank matrices $E = \frac{1}{\sqrt{T}} AB^T$, reducing memory from mn to $(m+n)r$ per layer (Sarkar et al., 2025), which inspires low-rank updates in our methods. Qiu et al. (2025) showed that ES can work at LLM scale, using layer-by-layer perturbation and evaluation. ES gives us a precursor for aligning evolutionary strategies with fitness hill climbing, increasingly advantaged as communication becomes a bottleneck. ES is a general optimizer, but cannot simply be naively applied to our setting without proper architecture design.

Post-training thicket noise perturbation also gives credence to ES for post-training (Gan and Isola, 2026), which found that pretraining creates solutions in dense spaces around the pretraining checkpoint such that random noise perturbation plus aggregation optimizes solutions well. These works motivate the intuition that ES in parameter space uses pretraining as a search heuristic, which motivates ALEC.

2.5 Memetic Evolution

Evolutionary communication units, known as "memes" in the Dawkins formalization (Dawkins, 1976), are like genes: stores of information that evolve by residing in humans. By competitive selection, ideas stay and spread among human memories to develop into "culture", a set of the most durable ideas (usually because they help their host to survive and spread the idea further). Their fitness depends not only on usefulness, but also on transmissibility, compatibility with host memory and attention, and robustness to mutation or reinterpretation. In memetic evolution as opposed to biological evolution, transmission and replication can be horizontal between organism lifetimes, and thus much faster than in biological evolution. Selection cycles may occur in weeks instead of years. This is a core motivator for recreating idea evolution in modern deployed systems.

Towards formalizing these notions, Weng et al. (2014) is a framework for predicting the behavior of idea-space latents in the form of posts on Twitter, showing that the entropy of clusters reached is the greatest predictor of eventual idea spread. This is one of the first attempts to formally analyze evolutionary idea dynamics in a deployment setting.

We now introduce Reward-Alignment in Latent Evolution (RALE) which measures alignment between latent evolution and host fitness. Formally, let Π denote the space of policies and let $S = \{s_1, \ldots, s_n\}$ denote the space of communicated latent strategies. We use an extractor $e : \Pi \to S$ to identify the latent strategy instantiated by a policy, so that $e(\pi) = s_i$ means policy π uses latent strategy s_i. $d : S \to \Pi$ is a decoder that maps a latent strategy to a representative policy realizing it.

Let P_t denote the population distribution over policies at round t. We define the frequency of latent strategy s_i as

$$x_i(t) = \Pr_{\pi \sim P_t}(e(\pi) = s_i), \qquad x_i(t) \geq 0, \qquad \sum_{i=1}^{n} x_i(t) = 1.$$

x The host fitness of latent strategy s_i in population state x is

$$F_i(x) = \mathbb{E}\left[\sum_{\tau=0}^{T} r_\tau \,\middle|\, e(\pi) = s_i,\, x \right].$$

We then introduce the mutated transmission operator

$$x_i(t+1) = \frac{1}{Z_t} \sum_j x_j(t) e^{\beta F_j(x(t))} Q_{ji}(x(t)),$$

where

$$Q_{ji}(x) = \Pr(e(\pi') = s_i \mid e(\pi) = s_j, x), \qquad Q_{ji}(x) \geq 0, \qquad \sum_i Q_{ji}(x) = 1.$$

Thus, $Q_{ji}(x)$ is the probability that an agent using latent strategy s_j before the learning phase uses latent strategy s_i afterward.

We define idea fitness as

$$R_i(x, Q) = \frac{1}{x_i} \sum_{k=1}^{n} x_k F_k(x) Q_{ki}(x), \qquad x_i > 0,$$

which measures the replication-weighted success of latent strategy s_i.

Our task is to learn Q s.t. $R_i(x, Q) = \phi(F_i(x))$ for some monotonic increasing ϕ, or that host fitness is aligned to idea fitness such that the more replicating ideas are also higher reward, such that a running system will reliably converge to high-fitness joint latent distributions. Thus we want to maximize alignment

$$A = -\mathbb{E}_x[\sum_i (R_i(x, Q) - \phi F_i(x))^2]$$

towards 0, at which point we consider a memetic evolution framework maximally "reward-aligned".

We can think of a memetic evolution framework thus in terms of how well it benefits its host. We think of high-A evolution frameworks as "parasitic", while low-A frameworks are "productive" and "growth-oriented".

3 Related Work

Bootstrapping of Memetics (Guttenberg and Rosa, 2021) uses an attention-based message-passing system to study the emergence of idea evolution and its interaction with weight dynamics, that is, how message exchange in genetically evolving agents leads to ideas. They observe little interaction between idea dynamics and underlying task performance. Cultural Evolution of Cooperation among LLM Agents (Vallinder and Hughes, 2024) places LLM agents in an iterated Donor Game in which each agent donates and the recipient receives $2\times$ the donated value; after several rounds, the top-performing agents survive and seed the next generation. Evolution through Large Models (**?**) uses large language models as mutation operators over code, and Altera's Project Sid (Ahn et al., 2024) studies long-horizon many-agent societies built through message exchange. Jaques et al. (2019) experiments with MARL agents that receive an intrinsic reward for exerting counterfactual causal influence over other agents' actions, which is abstractly related to prioritizing 'high-impact' ideas. They show this objective improves emergent communication using no centralized reward. Finally, LoRASA Zhang et al. (2025) applies low-rank adaptation to MARL policies, but not specifically to multi-agent communication itself. It trains a shared initial policy backbone and then learns a low-rank adapter on top of that backbone, connecting this design to the Eckart–Young–Mirsky theorem as a justification for compact low-rank adaptation, which is also relevant to ALEC.

4 Method

Attention-Guided Latent Evolution in Communication (ALEC) applies ES over a compact low-rank communication adapter layered on top of a pretrained attention-based MARL backbone with recurrent memory.

We begin from a pre-trained policy backbone consisting of an observation encoder, GRU task memory h, actor-critic trunk, and communication module. ALEC freezes this backbone and adapts only a small communication adapter ((to simulate a "post-training" setting). The adapter introduces rank-r low-rank deltas to the communication projections (Q, K, V, O) and maintains a persistent latent state z updated from the backbone memory h and communication context c.

OpenAI-ES is then applied over the flattened adapter genotype, with updates based on rank as a function of fitness. Inspired by MAPElites (Mouret and Clune, 2015), we also preserve and reevaluate a constant archive population.

the adapter ϕ, which determines individual messages and latent states. ϕ parameterizes low-rank perturbations to the frozen communication projections and persistent memory z. For a fixed genotype ϕ, the frozen backbone first produces observation encodings u_i^t and task memory states h_i^t. The

Algorithm 1 ALEC: Attention-Guided Latent Evolution via ES

Require: Frozen MARL backbone θ_{backbone}
Require: Adapter parameters ϕ_0, population size P, iterations T, noise scale σ, learning rate α

```
for t = 1 to T do
    for p = 1 to P do
        Sample noise ε_p ~ N(0, I)
        Form candidate adapter φ_p ← φ_{t-1} + σ ε_p
        Run rollouts with frozen backbone and candidate adapter φ_p
        Compute fitness R_p
    end for
    Shape fitness values {R_p}_{p=1}^P
    Update adapter
        φ_t ← φ_{t-1} + α · (1/P) Σ_{p=1}^P R_p ε_p
    Reevaluate elites and update archive
end for
return best adapter φ*
```

adapter then induces a communication process

$$q_i^t = q_{\text{base}}(h_i^t) + \Delta_Q^\phi([h_i^t; z_i^t]), \quad k_i^t = k_{\text{base}}(u_i^t) + \Delta_K^\phi(u_i^t), \quad v_i^t = v_{\text{base}}(u_i^t) + \Delta_V^\phi(u_i^t),$$

which produces a communication context c_i^t through attention, and updates the persistent memetic state via

$$z_i^{t+1} = (1 - \eta_i^t) z_i^t + \eta_i^t \tilde{z}_i^t,$$

with

$$\eta_i^t = \sigma\Big(\Delta_{\text{gate}}^\phi\big([z_i^t; h_i^t; c_i^t]\big)\Big), \qquad \tilde{z}_i^t = \tanh\Big(\Delta_{\text{cand}}^\phi\big([z_i^t; h_i^t; c_i^t]\big)\Big).$$

ALEC in the formal framework: In ALEC, the formal transmission operator is the communication-memory rule Q_ϕ induced by adapter parameters ϕ, while the meme states s_i are the persistent latent patterns propagated and stored through this rule in each z_i. x_i depends on a latent z_is frequency among agents in a given timestep, and Q_phi defines Q_{ij} and thus R_i and alignment A implicitly in its parametrization. See Table 5 for mapping between implemented objects in ALEC and formal objects in the reward-aligned memetic evolution framework.

Informally, we think of ALEC as meta-evolution, akin to biological evolution, to optimize idea evolution. In ALEC, the communication adapter defines the landscape along which "idea evolution" operates. In learning, we meta-evolve across adapters and generations, which then influences the evolution of which latents are communicated and held in persistent memory z. We can think of this as selection over 'memeplexes', or clusters of ideas which travel together (Blackmore, 1999), in which selection of individual ideas occurs.

5 Experiments

5.1 Predator-Prey in MPE2

Experimental Setup To study online adaptation, we start from a 200k-step MAPPO checkpoint and then apply a short post-training adaptation phase in MPE2 simple tag, in which predators collaborate to corner and contact prey, with reward based on minimum distances reached to prey. We compare ALEC against a MAPPO baseline under a matched environment-step budget, on a paired 128-episode evaluation at $N = 8$.

Both adaptation methods improved over the frozen checkpoint, but ALEC performed best overall. Mean reward improved from -449.20 for the frozen policy to -438.93 with MAPPO and -430.84 with ALEC. ALEC also achieved the best mean minimum-distance metric (0.407, versus 0.415 for RL and 0.428 for the frozen checkpoint). Communication interventions suggest that ALEC

Table 1: ALEC performance on MPE.

Method	Mean Reward	Mean min_dist	Seeds
Frozen	-449.20 ± 42.45	0.428 ± 0.047	3
MAPPO adapter	-438.93 ± 32.85	0.415 ± 0.035	3
ALEC	-430.84 ± 24.79	0.407 ± 0.027	3

also learned the most causally useful communication channel. Silencing communication caused the largest reward drop for ALEC (-25.85), compared with RL (-16.80) and the frozen checkpoint (-9.63). Message shifting was likewise more harmful for ALEC (-8.19) and RL (-6.44), while it improved the frozen checkpoint on average ($+5.98$), indicating that the frozen backbone had not yet learned a consistently useful communication channel.

5.2 Scaling in VMAS Transport and Discovery

Experimental Setup We evaluate post-training scaling across team sizes $N \in \{2, 4, 8, 12\}$ in VMAS Discovery and Transport across three seeds. In VMAS Discovery, N agents collaborate to cover targets while avoiding collisions. In VMAS Transport, N agents collaborate to push boxes into a goal zone. We initialize training from a 200k-step MAPPO backbone.

VMAS shows (noisy) parity in scaling between ALEC and MAPPO. Reward gain over each method's paired frozen checkpoint is positive for both methods at all tested N. In VMAS Discovery, ALEC is slightly stronger at $N = 4$ ($+0.1484$ vs. $+0.0911$), $N = 8$ ($+0.1198$ vs. $+0.1068$), and $N = 12$ ($+0.0885$ vs. $+0.0859$), while MAPPO is stronger only at $N = 2$ ($+0.1198$ vs. $+0.0469$). VMAS Transport is noisier: ALEC is better at $N = 2$ ($+0.0803$ vs. $+0.0463$), $N = 4$ ($+0.4376$ vs. $+0.2932$), and $N = 12$ ($+0.5993$ vs. $+0.4938$), while MAPPO is slightly better at $N = 8$ ($+0.4225$ vs. $+0.3383$).

5.3 Scaling in SMACv2

Experimental Setup We test scalng across agent team size $N \in \{2, 4, 8\}$ in SMACv2 Terran, holding the enemy team fixed at four units. We initialize from 500,000 steps of MAPPO, and train ALEC and MAPPO under a 30-minute wall-clock budget per seed. For ALEC we use (population 16, σ=0.05).

ALEC produces a positive mean improvement at every team size ($+0.04$, $+0.87$, $+0.21$) over the paired Pretraining baseline at N=2, 4, 8 respectively), and is the more consistent of the two methods across seeds. MAPPO is competitive on average but exhibits substantially higher variance: it strongly improves over the backbone at N=2 ($+0.79$), is neutral at N=4 ($+0.04$), and *regresses* at N=8 (-2.05, with one seed losing nearly 5 reward). Within the matched 30-minute budget, ALEC's per-seed work is roughly 73 ES generations on average (slightly more at N=2 where episodes are shorter), and MAPPO consumes on average $\sim$118k, $\sim$128k, and $\sim$131k transitions at N=2, 4, 8 respectively.

5.4 Meme-level F–R alignment (per-run clustering)

Experimental Setup To analyze alignment qualities of given latent evolution systems encoded by communication graphs, we log (z_t, z_{t+1}, r_t) in 25 deterministic evaluation episodes per seed for 3 seeds on Terran. Corresponding with discrete s_i, we discretize z with k-means with $K \in \{8, 12, \ldots, 128\}$ fitted separately for each (method, seed) pair, then estimate cluster occupancy x_k, host fitness $F_k = \mathbb{E}[Y \mid s^t{=}k]$, one-step Q, and replication weighting R. Concretely, each agent-step is labeled s_i^t by nearest k-means centroid in z_t; x_k is the fraction of such labels equal to k; the host-fitness target Y_i^t is the sum of the next H=10 per-step rewards along the same (episode, agent) trajectory, and Q_{kj} is the empirical frequency of $s_i^{t+1}{=}j$ among non-terminal steps with $s_i^t{=}k$; and $R_j{=}\frac{1}{x_j}\sum_k x_k F_k Q_{kj}$.

Table 2: Per-(method, seed) k-means, SMACv2 Terran. Means over 3 seeds; $\Delta\rho_w$ in same units. K is the requested cluster count (some bins may be dropped if occupancy $< 0.5\%$ at large K). Subscripts ^{AL} and ^{MP} denote ALEC and MAPPO.

Nv4	K	ρ_w^{AL}	$\Delta\rho_w^{\mathrm{AL}}$	$A_{\mathrm{iso}}^{\mathrm{AL}}$	ρ_w^{MP}	$\Delta\rho_w^{\mathrm{MP}}$	$A_{\mathrm{iso}}^{\mathrm{MP}}$
2	8	0.800	+0.19	0.039	0.864	+0.46	0.075
2	12	0.886	+0.19	0.029	0.879	+0.35	0.074
2	16	0.904	+0.18	0.030	0.882	+0.30	0.072
2	20	0.884	+0.14	0.034	0.851	+0.22	0.081
2	24	0.883	+0.12	0.032	0.853	+0.22	0.086
2	32	0.877	+0.10	0.035	0.866	+0.20	0.082
2	64	0.879	+0.07	0.036	0.878	+0.18	0.082
2	128	0.877	+0.05	0.036	0.869	+0.23	0.074
4	8	0.907	+0.32	0.110	0.920	+0.41	0.062
4	12	0.908	+0.22	0.115	0.931	+0.31	0.054
4	16	0.935	+0.20	0.104	0.939	+0.24	0.095
4	20	0.926	+0.18	0.113	0.948	+0.23	0.100
4	24	0.935	+0.14	0.116	0.945	+0.18	0.111
4	32	0.932	+0.12	0.125	0.942	+0.16	0.113
4	64	0.941	+0.09	0.110	0.957	+0.15	0.108
4	128	0.942	+0.07	0.107	0.944	+0.14	0.118
8	8	0.961	+0.59	0.145	0.855	+0.68	0.084
8	12	0.955	+0.43	0.137	0.888	+0.58	0.084
8	16	0.959	+0.37	0.157	0.908	+0.58	0.084
8	20	0.970	+0.35	0.138	0.907	+0.50	0.082
8	24	0.968	+0.32	0.177	0.921	+0.52	0.086
8	32	0.971	+0.29	0.186	0.936	+0.49	0.124
8	64	0.975	+0.23	0.238	0.930	+0.42	0.247
8	128	0.965	+0.20	0.304	0.935	+0.41	0.282

Metrics. *Weighted Spearman* $\rho_w(F, R)$: rank correlation between vectors (F_k) and (R_k) with weights (x_k), measuring whether higher-host-fitness clusters also rank higher in replication weighting R.

Isotonic mismatch $A_{\mathrm{iso}} = \sum_k x_k \big(R_k - \phi(F_k)\big)^2$ where ϕ is the isotonic (monotone) regression of R on F with weights x_k; $A_{\mathrm{iso}} = 0$ iff $F \mapsto R$ is weakly monotone across kept clusters, and larger values indicate systematic non-monotonicity.

Shuffle null: we permute the per-step target Y across agent-steps, recompute F and R with fixed (s^t, s^{t+1}), and average ρ_w over 50 shuffles; $\Delta\rho_w \equiv \rho_w - \rho_w^{\mathrm{shuf}}$ is the gap above this structural baseline (tight Q and geometry can make both numbers large, so $\Delta\rho_w$ is the more informative alignment signal for comparing methods).

Results In Table 2, both methods achieve high ρ_w across K and team sizes, but $\Delta\rho_w$ (above shuffle) more often favours MAPPO while A_{iso} is typically lower (better) for ALEC at N=2 and more mixed for N=4 and 8; at very large K the partition is an over-segmentation of z and A_{iso} can grow, so K is a resolution choice, not a unique meme count. We do not see a clear overall winner on all three of ρ_w, $\Delta\rho_w$, and A_{iso}, so we report the tables as evidence that *both* phase-2 adapters yield reward-linked meme statistics under a fair per-run codebook, with differences concentrated in shuffle-corrected rank alignment and small-team isotonicity.

5.5 Ablations

Persistent memory helps performance On partial-observation MPE simple spread, we compare memory-less vs memory ALEC. Memory is the most stable option as team size grows. It keeps final mean coverage distance low at every tested N (0.678 ± 0.059, 0.695 ± 0.180, 0.613 ± 0.065 at $N = 3, 5, 8$; $n = 6$ each), whereas the memory-free baseline rises to 9.815 ± 9.203, 16.325 ± 12.717, and 13.425 ± 12.945. The SGD-trained memory+communication variant is less stable still, reaching 10.225 ± 9.575, 38.275 ± 16.793, and 26.127 ± 23.589 ($n = 6, 6, 3$). Final rewards show the same

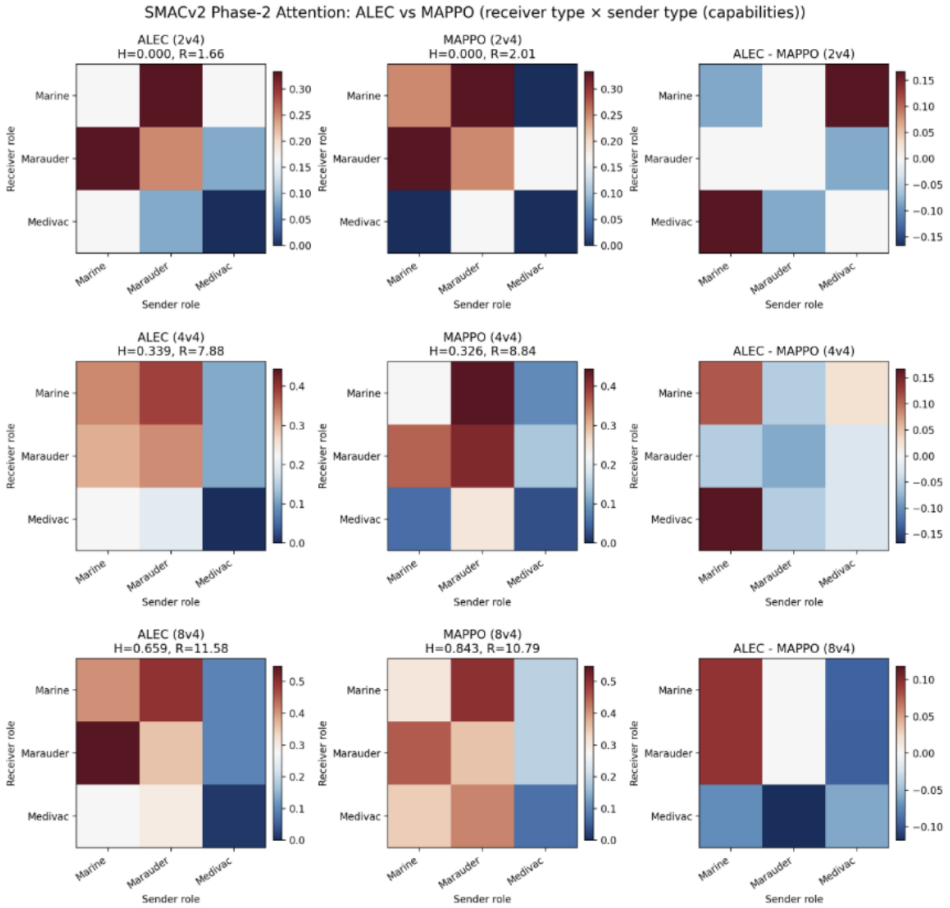

Figure 1: Visualization of attention graphs of ALEC, MAPPO, and Δ between them on SMACv2 by Terran role for $N = \{2, 4, 8\}$v4. Across all N, the combat-oriented roles (Marine, Marauder) seem to attend strongly to each other, whereas the Medivac is largely ignored. ALEC in comparison to MAPPO seems to prioritze medivac attention to others more.

pattern: memory-only stays between -646.90 ± 31.55 and -1200.81 ± 80.29, while the baseline and memory+communication variants are both worse and far more variable.

ALEC is more wall-clock efficient than MAPPO On a single Apple M3 Pro MacBook Pro (12 CPU cores, 36 GB unified memory; CPU execution), ALEC is faster overall than matched RL adaptation, though not in every VMAS Transport setting. Across the 27 paired runs in the current artifact set, ALEC averaged 3601.0 ± 654.6 seconds versus 4493.5 ± 599.0 for RL, and ALEC finished faster in 22/27 pairings.

Low-rank state updates appear slightly better and more stable than dense alternatives. On VMAS Discovery at $N = 12$, the standard low-rank z update achieves reward gain $+0.0885 \pm 0.0138$ and improves in 3/3 seeds, compared with $+0.0755 \pm 0.0478$ and 2/3 seeds for dense updates, and $+0.0547 \pm 0.0430$ and 2/3 seeds with z disabled. The low-rank variant is also materially cheaper wall-clock.

Starting from an adapter and not from uninitialized is motivated by OpenAI ES estimates the gradient via random directional perturbations. The variance of that estimate scales roughly linearly with parameter count (you need $O(d/N)$ population to keep the gradient SNR fixed). The full model is 236k params; the typical ALEC adapter is $1\breve{}10k$. So the same population size does 100× more "useful work per generation" on the adapter.

Backbone vs. from-scratch search on SMACv2 4v4 Terran. We compare initializing ES from pretraining using three time-matched adaptation methods on the SMACv2 4v4 Terran scenario, all granted the same wall-clock budget per seed on an NVIDIA RTX3090 with 32 vCPUs. We compare wall-clock-matched whole-model OpenAI-ES from scratch to MAPPo and ALEC from a 500k-step pretrained backbone (included in comparable wall-clock time) as an ablation on pretraining. Non-pretrained ES consumes roughly $7\times$ more transitions than the MAPPO adapter ($\sim$956k vs. $\sim$128k) in the same time, yet collapses to less than 60% of the pretrained baseline reward. ALEC recovers the largest gain over the paired pretrained checkpoint ($+0.87$ mean reward), MAPPO is essentially neutral ($+0.04$), and whole-model ES regresses by -5.01. The result indicates that under a fixed wall-clock budget, the marginal value of pretraining is not recovered in 30 minutes of from-scratch search, and that adapter-only search on top of a pretrained backbone is the more sample-efficient method.

Pooled-seed, per-method clustering Instead of a fresh k-means for every seed, we fit *one* codebook per method by pooling (subsampled) z from all 3 seeds, then assign every probe. This matches cluster identities across seeds within a method, but can interact badly with differing latent scales between ALEC and a narrow MAPPO cloud if one later pooled across methods (we do *not* do that here). The columns val AL / val MP are $n_{\text{valid}}/n_{\text{total}}$ seeds for which ρ_w and A_{iso} are defined (need ≥ 3 kept clusters with sufficient mass).

6 Conclusion and Future Work

We introduced a formalism for reward-aligned latent evolution in multi-agent communication and instantiated it in two methods: LA-PPO and ALEC. LA-PPO makes is a simple preliminary implementation of the framework, while ALEC provides a simpler and more scalable post-training adaptation framework using ES over a compact communication-and-memory interface.

Empirically, LA-PPO demonstrates that latent evolutionary dynamics can be embedded into modern MARL systems and can improve exploratory capacity, though with instability relative to strong communication baselines. ALEC shows more promising scalability and alignment properties: under matched adaptation budgets, it is often competitive with or better than RL adaptation, and its search-time selection signal is better aligned with held-out performance.

Cultural evolution proceeds on a much faster timescale than genetic evolution and cumulative cultural evolution is widely argued to underlie much of humanity's adaptive and technological success (Richerson and Boyd, 2008; Mesoudi and Thornton, 2018). Developing on findings from LA-PPO and ALEC, can we design memetic evolution methods that scale to pretrained model communication latents and larger agent populations?

Taking inspiration from quality-diversity (Mouret and Clune, 2015; Pugh et al., 2016) and open-endedness theory (Hughes et al., 2024), can we study and improve the open-endedness properties of evolution over communication? For example, future work may skew more toward open-endedness/evolutionary-focused goals where the aim is to continue searching and producing novelty in the evolutionary idea search, perhaps by including quality-diversity or self-play elements.

Inspired by Andreas et al. (2017), how can we mechanistically interpret inter-agent communication dynamics and changes in latents over time to ensure continual model alignment (Patel et al., 2021; Carmeli et al., 2024)? Humans became much more than our biological networks would suggest, purely by the gradual exchange and selective persistence of ideas. Similarly, models may be aligned at deployment time, but morph over time via multi-agent communication into something unpredictable. Safety, alignment, and interpretability research ought increasingly be concerned with deciphering and understanding the meaning and dynamics of multi-agent communication.

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Table 3: VMAS Discovery mean reward by team size N. For each method, the paired frozen checkpoint mean is reported alongside the adapted mean. Values are mean $\pm$ standard error; the final column reports the seed count.

N	ALEC Base	ALEC	MAPPO Base	MAPPO	Seeds
2	0.5443 ± 0.1134	0.5911 ± 0.0290	0.5000 ± 0.0597	0.6198 ± 0.1090	3
4	0.7630 ± 0.0104	0.9115 ± 0.0104	0.7760 ± 0.0248	0.8672 ± 0.0325	3
8	1.2396 ± 0.0648	1.3594 ± 0.0430	1.2839 ± 0.0456	1.3906 ± 0.0835	3
12	1.4792 ± 0.0114	1.5677 ± 0.0213	1.4661 ± 0.0293	1.5521 ± 0.0094	3

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A Implementation Details

A.1 LA-PPO Implementation Details

In the reported MPE2 Simple Tag experiments, we use continuous actions, 3 predator agents, 1 prey agent, 2 obstacles, maximum episode length 25, and 8 parallel environments. The main step-matched comparison uses a budget of 200,000 environment steps. PPO uses Adam with learning rate 3×10^{-4}, discount $\gamma = 0.95$, GAE parameter $\lambda = 0.95$, clip coefficient 0.2, entropy coefficient 0.01, value-loss coefficient 0.5, maximum gradient norm 0.5, 4 update epochs per rollout, batch size 64, and rollout horizon 128. The encoder uses hidden sizes $[128, 128]$ and latent dimension 64; the policy and value heads are shallow MLPs on top of this latent. We apply linear learning-rate annealing to zero over the estimated number of PPO updates.

Latent alignment is applied every 10 learning steps through an asymmetric distillation loss between agent encoders. We use distillation weight 0.1, distillation batch size 64, and per-agent observation-buffer size 512, with an all-to-all ordered-pair distillation topology. Evaluation during training uses a rolling opponent pool with evaluation every 20,000 environment steps, Elo computed from 5 episodes per checkpoint, pool size 5, and cross-play matrices every 100,000 steps.

A.2 ALEC Implementation Details

The adapter uses a persistent latent state $z \in \mathbb{R}^{16}$ and rank-4 low-rank deltas on (Q, K, V, O). By default, the z update is also low-rank; dense-update and no-z variants are used only in ablations. Episodes are 200 steps long. OpenAI-ES is applied to the flattened adapter genotype with centered-rank fitness shaping and antithetic Gaussian perturbations, using noise scale $\sigma = 0.02$ and learning rate 0.005. Cheap evaluation uses 4 episodes per candidate. In the main VMAS $N = 12$ runs, the `linear_a` regime uses population size 16 and 20 generations, while `fixed_a` uses population size 10 and 14 generations. The stronger runs reevaluate the top 4 candidates on 32 episodes, keep an archive of size 8, and perform final held-out selection on 64 episodes.

The matched RL baseline trains the same adapter with Adam using learning rate 3×10^{-4}, discount $\gamma = 0.99$, GAE $\lambda = 0.95$, entropy coefficient 0.01, maximum gradient norm 10.0, and rollout length 256 under matched transition budgets. Evaluation is deterministic and paired, and includes message-silencing and message-shifting interventions. In the stronger runs, final selection may also use a baseline guard that keeps the frozen checkpoint if the adapted candidate fails on held-out evaluation.

Table 4: VMAS Transport mean reward by team size N. For each method, the paired frozen checkpoint mean is reported alongside the adapted mean. Values are mean $\pm$ standard error; the final column reports the seed count.

N	ALEC Base	ALEC	MAPPO Base	MAPPO	Seeds
2	-0.0319 ± 0.0302	0.0484 ± 0.0382	0.0059 ± 0.0092	0.0522 ± 0.0341	3
4	-0.3133 ± 0.0645	0.1243 ± 0.0342	-0.2018 ± 0.1987	0.0914 ± 0.0355	3
8	0.0837 ± 0.1550	0.4220 ± 0.1246	0.0548 ± 0.0628	0.4773 ± 0.1987	3
12	-0.4621 ± 0.4255	0.1371 ± 0.1550	-0.2727 ± 0.1599	0.2211 ± 0.2738	3

Table 5: Mapping from the formal memetic framework to ALEC (phase 2, frozen MAPPO backbone). The learned object is the adapter genotype ϕ; post-hoc RALE analysis discretizes z to compare F and R on the induced rollouts.

Formal object	ALEC implementation
Meme state s_i	At run time each agent has a *continuous* memetic state $z_i^t \in \mathbb{R}^{d_z}$ from the `MemeticStateCell`, with cross-agent message context c_i^t; the adapter is stacked on a frozen `FrozenAttentionHUActorBackbone` checkpoint. For RALE, discrete labels s_i^t are *post-hoc*—k-means on logged z_i^t (with optional $[z; c]$ in control experiments)—so the formal discrete meme index k is the cluster id, not an output head of the policy.
Population frequency x_i / x_k	x_k is the fraction of agent–steps in cluster k on evaluation rollouts under a *fixed* trained adapter, $x_k = \frac{1}{M}\sum_i \mathbf{1}[s_i^t{=}k]$. This is the measured analogue of type frequency, not a quantity differentiated inside ES.
Host fitness F_i / F_k	Outer-loop ES fitness is scalar return (mean per-episode reward, with optional comm-gain terms in the selection script). For the RALE table, $F_k = \mathbb{E}[Y_i^t \mid s_i^t{=}k]$ with Y_i^t a short-horizon host target, e.g. the sum of the next H per-step rewards on the same (episode, agent) trajectory.
Replicative fitness R_i / R_k	In RALE, $R_j = \frac{1}{x_j}\sum_k x_k F_k Q_{kj}$: replication-weighted fitness from the empirical cluster–conditional F and the *empirical* one-step matrix Q on discretized s; it is not an extra parametric "reproduction" head.
Transmission / dynamics (two meanings of Q)	*Learned dynamics:* low-rank deltas $(\Delta_Q, \Delta_K, \Delta_V, \Delta_O)$ on the communication Q/K/V/O paths and the gated state update $z_{t+1}{=}f(z_t, h_t, c_t)$ in the adapter. *Empirical RALE Q_{kj}:* $\Pr(s^{t+1}{=}j \mid s^t{=}k)$ on the same discretized trajectories, used only to build R; do not identify this with the neural attention Q–projection weights.
Meta-level mutation (algorithmic)	OpenAIES on the flattened adapter: $\phi' = \phi + \sigma\epsilon$, *antithetic* sampling, gradient from centered rank transform of population fitness, learning rate η. The phase-2 runner can optionally re-evaluate top candidates and keep an elite archive; that outer loop over ϕ is not the in-episode z update.

A.3 VMAS Scaling Results

A.4 RALE ALEC Component Mapping

A.5 ALEC wall-clock comparison vs MAPPO

A.6 ALEC Dense vs Low-rank z Ablation

Table 6: Matched wall-clock by domain. Values are mean $\pm$ standard error in seconds over paired runs on a single Apple M3 Pro MacBook Pro (12 CPU cores, 36 GB unified memory) using CPU execution.

Setting	ALEC Time (s)	MAPPO Time (s)	Pairs	ALEC Faster
MPE	1156.0 ± 6.5	1312.9 ± 17.0	3	3/3
VMAS Discovery	2792.0 ± 478.8	5359.0 ± 1127.6	12	12/12
VMAS Transport	5021.4 ± 1292.5	4423.2 ± 581.4	12	7/12
Overall	3601.0 ± 654.6	4493.5 ± 599.0	27	22/27

Table 7: VMAS Discovery $N = 12$ state-update ablation for ALEC. Values are mean $\pm$ standard error over seeds.

Variant	Baseline Reward	Final Reward	Reward Gain	Time (s)	Improved	Seeds
with z, low-rank update	1.4792 ± 0.0114	1.5677 ± 0.0213	0.0885 ± 0.0138	5267.8 ± 170.8	3/3	3
without z	1.5078 ± 0.0798	1.5625 ± 0.0413	0.0547 ± 0.0430	7920.8 ± 1026.8	2/3	3
with z, dense update	1.5234 ± 0.0502	1.5990 ± 0.0026	0.0755 ± 0.0478	16163.8 ± 634.0	2/3	3

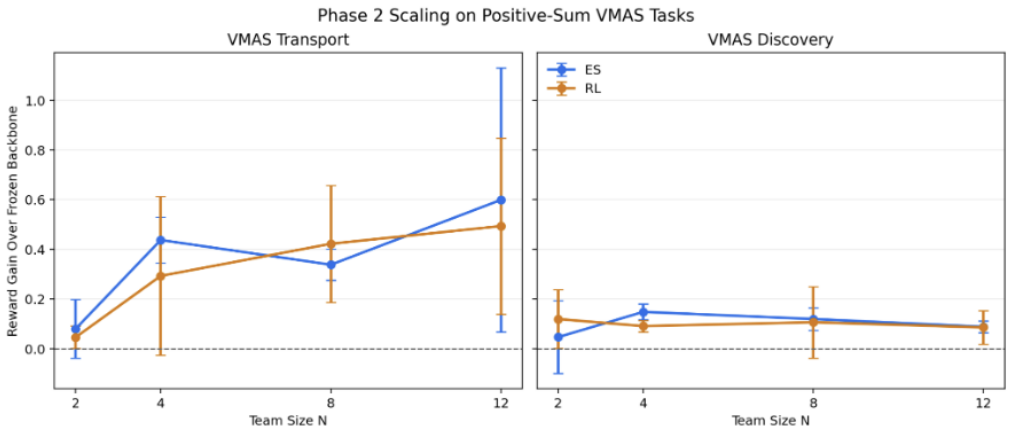

Figure 2: ALEC scaling on VMAS Discovery and Transport. Gains are computed relative to each method's paired frozen checkpoint.

Method	Pretraining	Adapt. method	Pretraining budget	Pretraining work	Reward (mean
Pretraining backbone	500k MAPPO steps	—	—	—	11.33 ± 1
ALEC (Arm A)	500k MAPPO steps	adapter ES (OpenAI-ES)	1800 s	$\sim$73 generations	12.20 ± 1
MAPPO (Arm B)	500k MAPPO steps	adapter PPO	1800 s	$\sim$128k transitions	11.36 ± 0
Whole-model ES (Arm C)	*none* (random init)	whole-model OpenAI-ES	1800 s	$\sim$956k transitions	6.32 ± 0

Table 8: Adaptation on SMACv2 4v4 Terran with a 30-minute matched wall-clock budget. Mean $\pm$ standard error across n=3 seeds (seeds 42, 43, 44). Pretrained backbones are trained for 500,000 environment steps of MAPPO with the attention_hu_actor architecture; the same checkpoint is reused as the pretrained baseline reference and as the frozen backbone for Arms A and B. Arm C re-initializes the backbone randomly and runs whole-model OpenAI-ES with population 16 and σ=0.05.

N vs. 4	Pretraining base	ALEC: mean $\pm$ SE (Δ)	MAPPO: mean $\pm$ SE (Δ)	ALEC gens. / MAPPO trans.
2	2.33 ± 0.16	2.38 ± 0.07 ($+0.04$)	3.12 ± 0.11 ($+0.79$)	$\sim$122 / $\sim$118k
4	11.33 ± 1.40	12.20 ± 1.64 ($+0.87$)	11.36 ± 0.75 ($+0.04$)	$\sim$73 / $\sim$128k
8	22.64 ± 2.56	22.86 ± 2.42 ($+0.22$)	20.59 ± 4.15 (-2.05)	$\sim$73 / $\sim$131k

Table 9: ALEC vs. MAPPO team-scaling on SMACv2 Terran with the enemy team fixed at four units. Δs show mean difference compared to baseline mean.

Table 10: Partial-observation MPE simple_spread scaling ablation. Values are mean $\pm$ standard error over seeds; lower mean_dist is better. The final column reports seeds as baseline / memory-only / memory+communication. The memory+communication row at $N = 8$ uses 3 seeds; all other entries use 6.

N	Baseline Reward	Memory-Only Reward	Memory+Comm Reward	Baseline mean_dist	Memory-Only mean_dist	Memory+Comm mean_dist	Seeds
3	-3257.73 ± 2634.42	-646.89 ± 31.55	-3997.40 ± 3371.98	9.815 ± 9.203	0.678 ± 0.059	10.225 ± 9.575	6/6/6
5	-8577.72 ± 6199.96	-868.27 ± 158.39	-18768.44 ± 7951.74	16.325 ± 12.717	0.695 ± 0.180	38.275 ± 16.793	6/6/6
8	-11124.43 ± 10070.12	-1200.81 ± 80.29	-21864.15 ± 18150.99	13.425 ± 12.945	0.613 ± 0.065	26.127 ± 23.589	6/6/3

Table 11: Same metrics as Table 2 with k-means on **pooled** z per method (ALEC vs. MAPPO). Compare to Table 2 to see how much conclusions depend on cluster scope. Subscripts ^{AL} and ^{MP} denote ALEC and MAPPO.

Nv4	K	val AL	val MP	ρ_w^{AL}	$\Delta\rho_w^{\mathrm{AL}}$	$A_{\mathrm{iso}}^{\mathrm{AL}}$	ρ_w^{MP}	$\Delta\rho_w^{\mathrm{MP}}$	$A_{\mathrm{iso}}^{\mathrm{MP}}$
2	8	3/3	3/3	0.863	+0.54	0.023	0.164	+0.24	0.082
2	12	3/3	3/3	0.862	+0.45	0.031	0.773	+0.56	0.066
2	16	3/3	3/3	0.744	+0.25	0.030	0.785	+0.49	0.075
2	20	3/3	3/3	0.855	+0.25	0.029	0.813	+0.42	0.076
2	24	3/3	3/3	0.859	+0.32	0.028	0.836	+0.38	0.081
2	32	3/3	3/3	0.878	+0.22	0.032	0.872	+0.37	0.073
2	64	3/3	3/3	0.895	+0.16	0.031	0.887	+0.27	0.077
2	128	3/3	3/3	0.887	+0.07	0.033	0.868	+0.20	0.081
4	8	2/3	3/3	0.966	+0.60	0.039	0.896	+0.64	0.034
4	12	3/3	3/3	0.767	+0.25	0.055	0.922	+0.65	0.024
4	16	3/3	3/3	0.799	+0.27	0.098	0.949	+0.58	0.057
4	20	3/3	3/3	0.955	+0.47	0.073	0.814	+0.32	0.063
4	24	3/3	3/3	0.950	+0.46	0.096	0.898	+0.32	0.062
4	32	3/3	3/3	0.919	+0.36	0.106	0.872	+0.21	0.059
4	64	3/3	3/3	0.909	+0.19	0.112	0.940	+0.22	0.103
4	128	3/3	3/3	0.940	+0.13	0.116	0.947	+0.18	0.111
8	8	3/3	3/3	0.450	+0.29	0.045	1.000	+1.02	0.000
8	12	3/3	3/3	0.433	−0.02	0.040	0.856	+0.77	0.074
8	16	3/3	3/3	0.960	+0.57	0.160	0.809	+0.65	0.060
8	20	3/3	3/3	0.953	+0.53	0.168	0.893	+0.71	0.063
8	24	3/3	3/3	0.954	+0.51	0.156	0.876	+0.66	0.064
8	32	3/3	3/3	0.941	+0.48	0.155	0.887	+0.58	0.032
8	64	3/3	3/3	0.968	+0.34	0.147	0.941	+0.52	0.089
8	128	3/3	3/3	0.963	+0.26	0.243	0.944	+0.47	0.200